

Experiment No.8

Aim: To study and Implement Storage as a Service using Own Cloud/ AWS S3, Glaciers/Azure Storage.

Objectives:

- To Develop a theoretical understanding of Storage as a Service (SaaS) principles, emphasizing features such as scalability, durability, and cost-effective storage solutions.
- Apply practical knowledge to configure and utilize AWS S3 and Glacier for scalable, reliable, and cost-efficient storage solutions.

Outcomes:

- Demonstrate proficiency in utilizing AWS S3 and Glacier to store, retrieve, and manage data in a scalable and secure manner.
- Apply knowledge in implementing Storage as a Service, understanding how to optimize storage solutions based on data access patterns and lifecycle requirements.

Software Requirements: Windows/Ubuntu OS, AWS Account.

Theory:

Storage as a Service (SaaS) involves the provision of scalable and on-demand storage solutions in a cloud environment. Amazon S3 is an object storage service designed to store and retrieve any amount of data from anywhere on the web. It offers high durability, availability, and scalability. Amazon Glacier is a low-cost storage service for archiving and long-term backup, optimized for infrequently accessed data.

Amazon S3 provides a simple web interface to store and retrieve any amount of data. It organizes data into "buckets," and each object within a bucket is assigned a unique key. S3 offers different storage classes, including Standard, Intelligent-Tiering, Standard-IA, Glacier, and Deep Archive, each catering to specific use cases and access patterns.

Amazon Glacier is designed for archiving purposes, offering low-cost storage for data that is rarely accessed. It provides configurable retrieval times, allowing users to optimize costs based on the urgency of data retrieval.

Key Components:

1. **Cloud Storage Provider:** Offers storage infrastructure and services, managing the underlying hardware and software.
2. **User/Application:** Interfaces with the cloud storage service via APIs, web consoles, or other tools.
3. **Data Objects:** Files, images, videos, or any other digital content stored in the cloud.
4. **Storage Buckets:** Logical containers for organizing and managing data objects.
5. **Metadata:** Information about data objects, such as file names, sizes, creation dates, and access permissions.
6. **Access Control:** Mechanisms to regulate who can access data and what actions they can perform.
7. **Data Replication:** Copies of data stored across multiple locations for redundancy and availability.

How It Works:

1. **Data Upload:** User/application uploads data objects to the cloud storage service.
2. **Object Storage:** Data objects are stored as discrete units, often with their own metadata and access controls.
3. **Data Access:** User/application can retrieve, modify, or delete data objects as needed.
4. **Data Management:** Cloud storage provider handles data maintenance, including replication, backup, and disaster recovery.

Stepwise Procedure to Implement PaaS using AWS Elastic Beanstalk:

Step 1: Sign into AWS Console

- Log in to the AWS Management Console using your AWS account credentials.

Step 2: Access S3 Service

- Navigate to the S3 service in the AWS Console.

Step 3: Create S3 Bucket

- Create an S3 bucket to store objects.
- Configure bucket settings, including region and versioning.

Step 4: Upload Objects to S3

- Upload files or objects to the S3 bucket.
- Configure access permissions for the objects.

Step 5: Set Object Lifecycle Policies

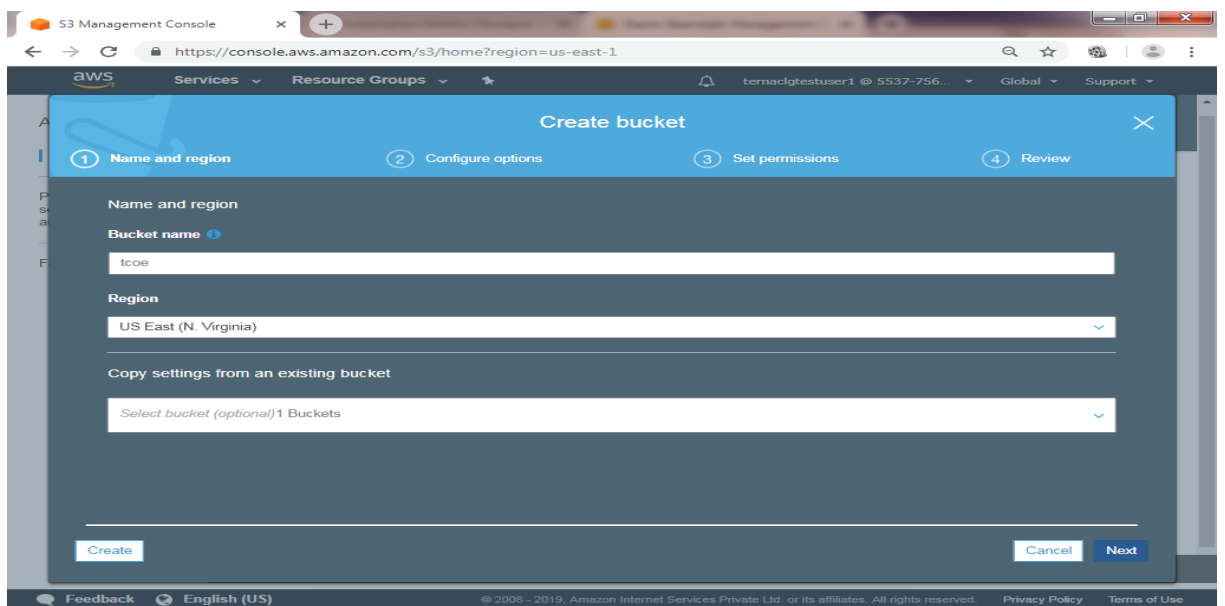
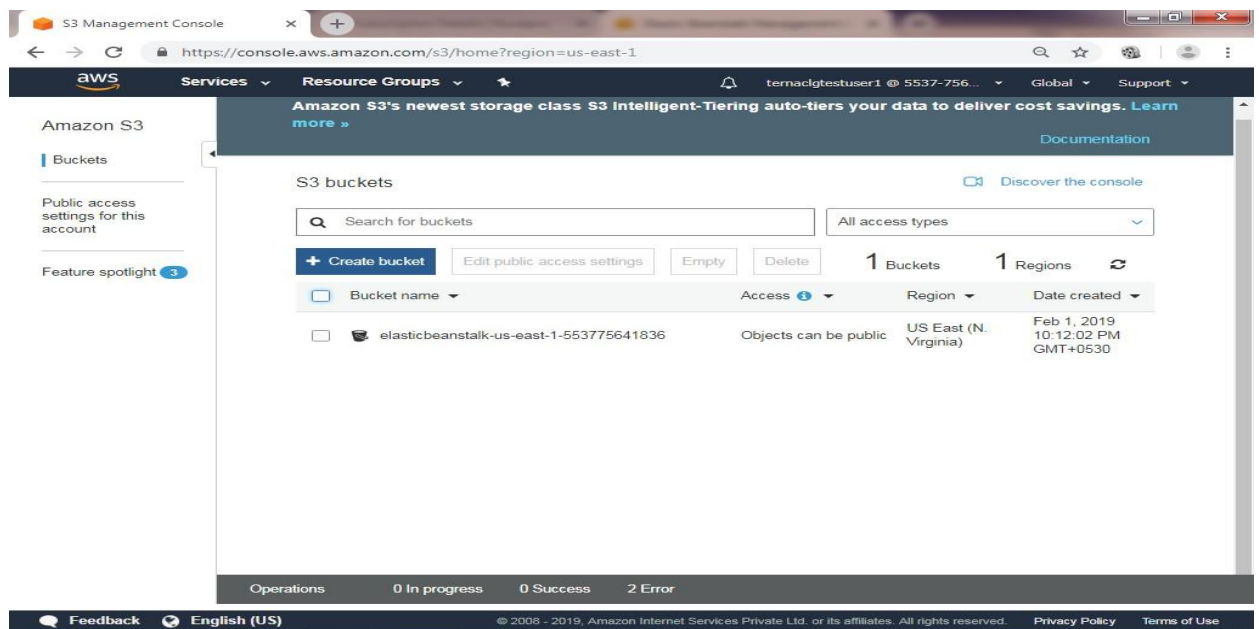
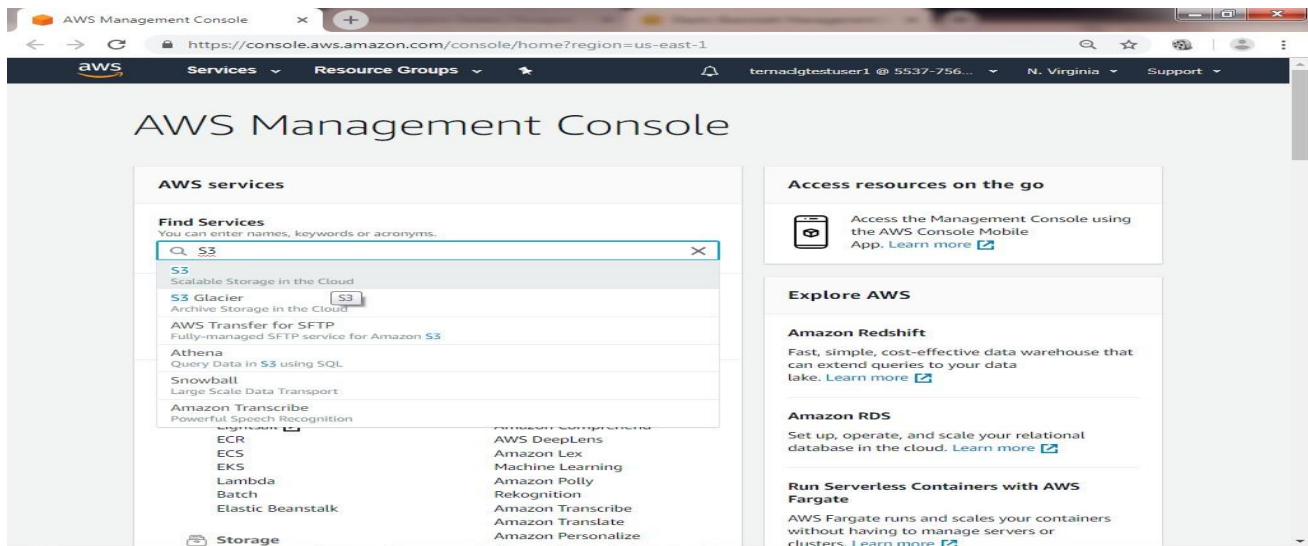
- Define object lifecycle policies based on access patterns.
- Configure object transitions to different storage classes, such as Glacier.

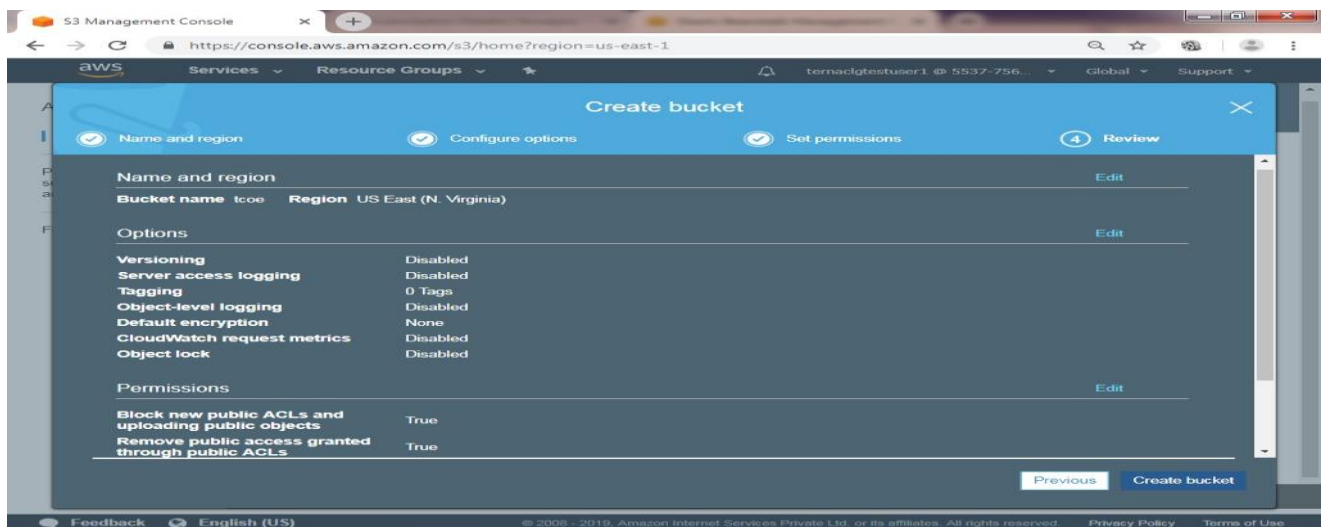
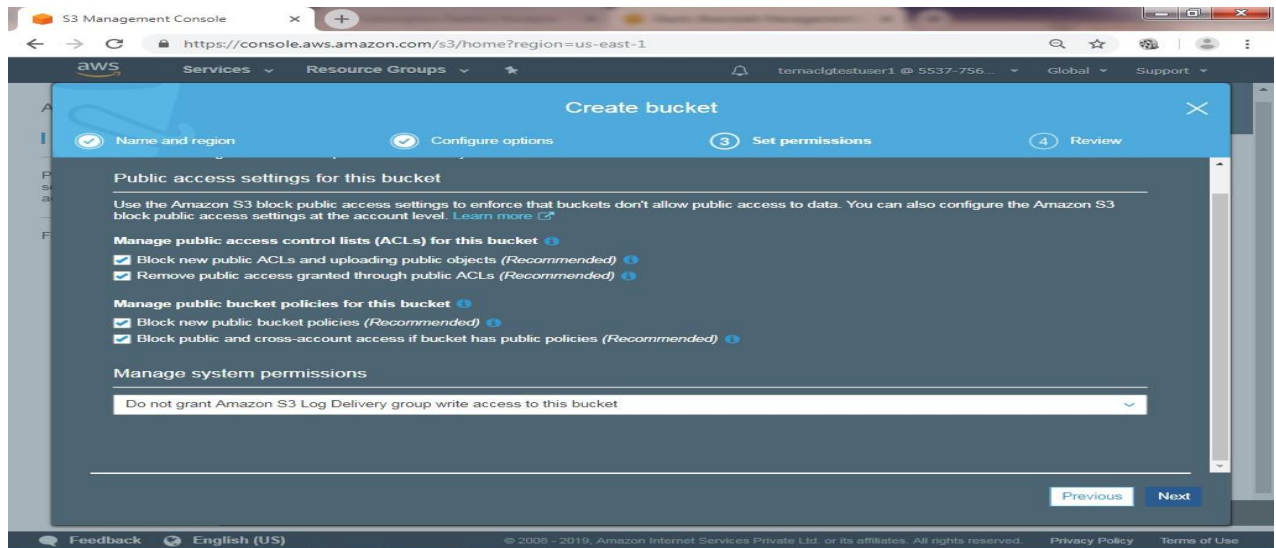
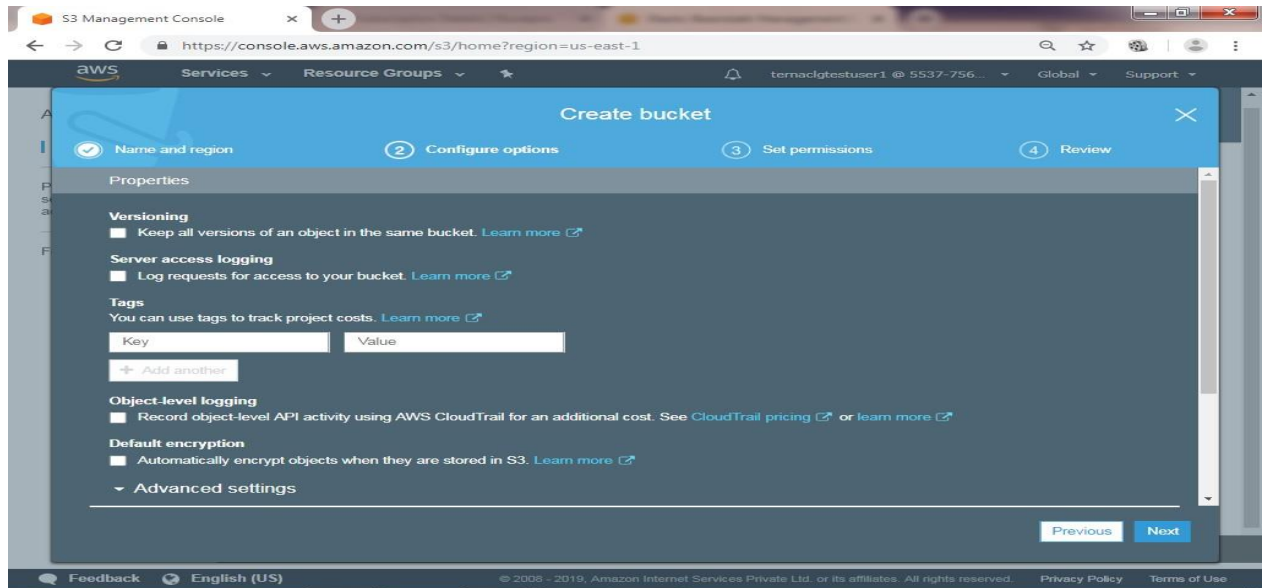
Step 6: Access Glacier Service

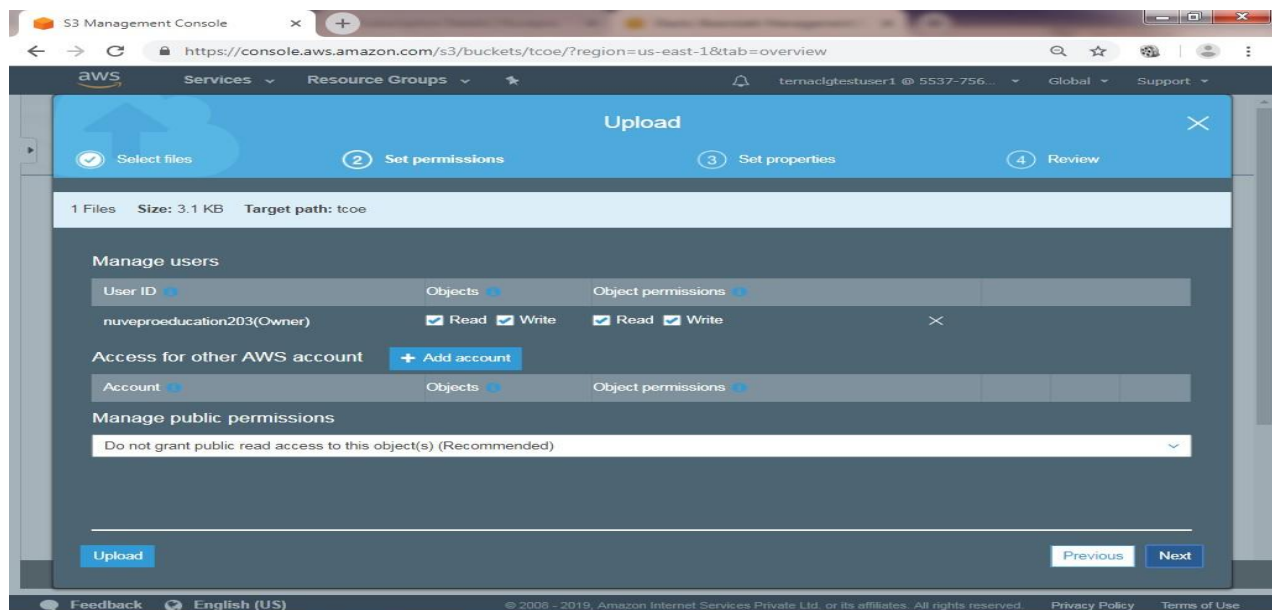
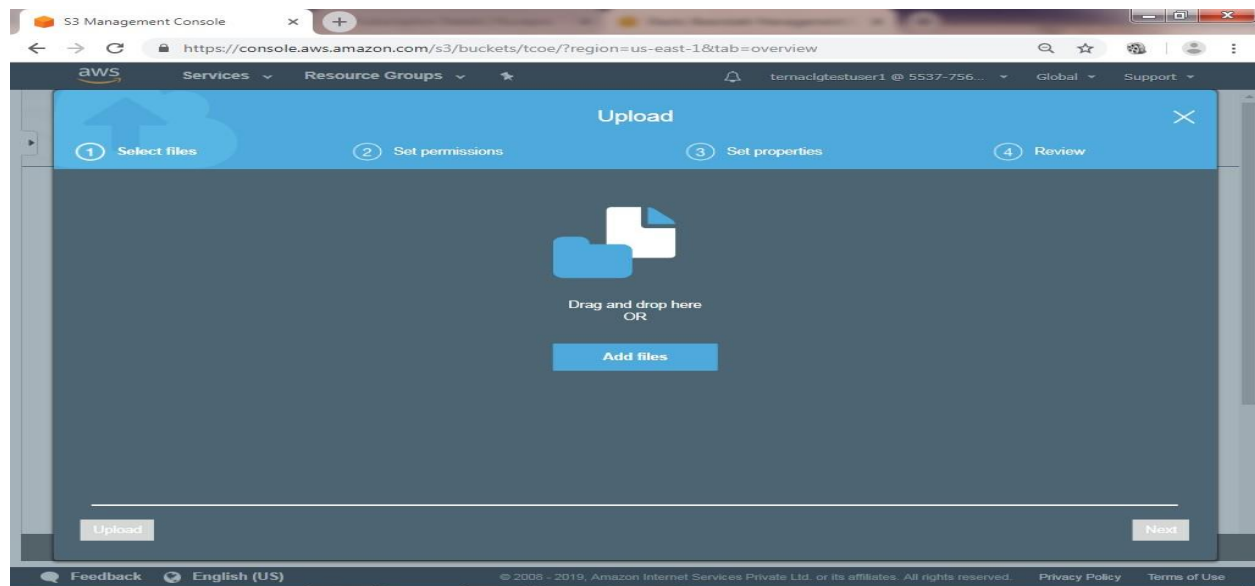
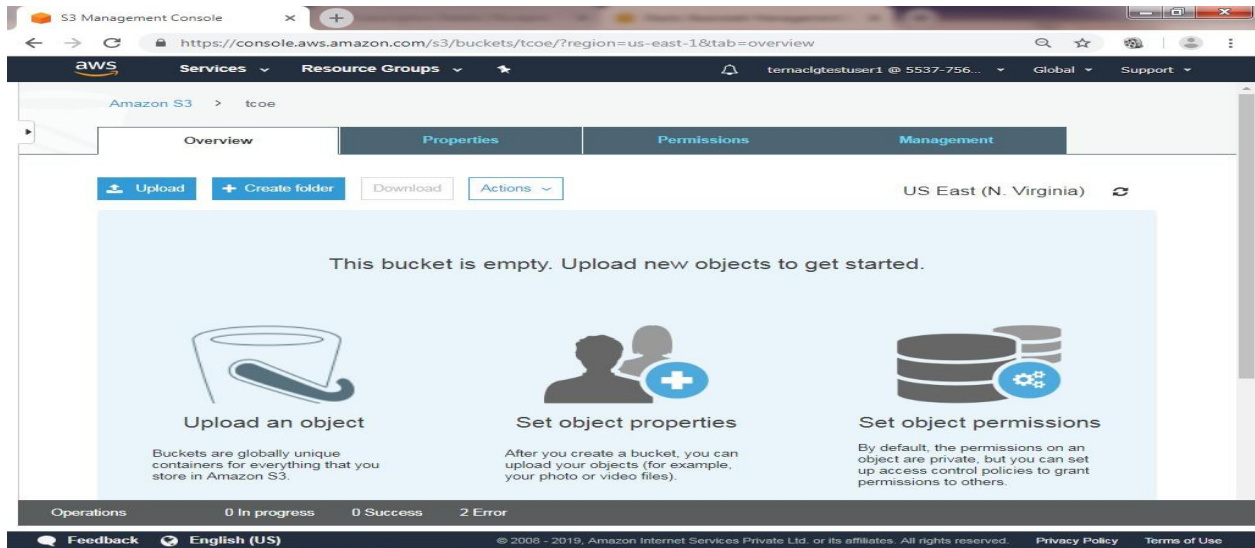
- Navigate to the Glacier service in the AWS Console.
- Create a Glacier vault for archiving.

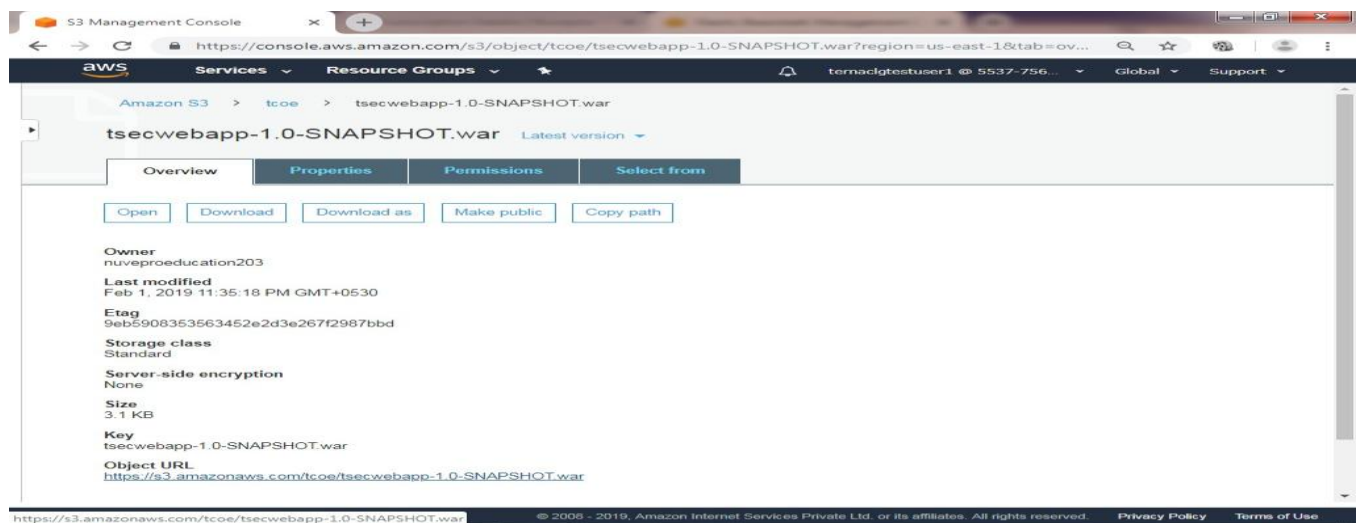
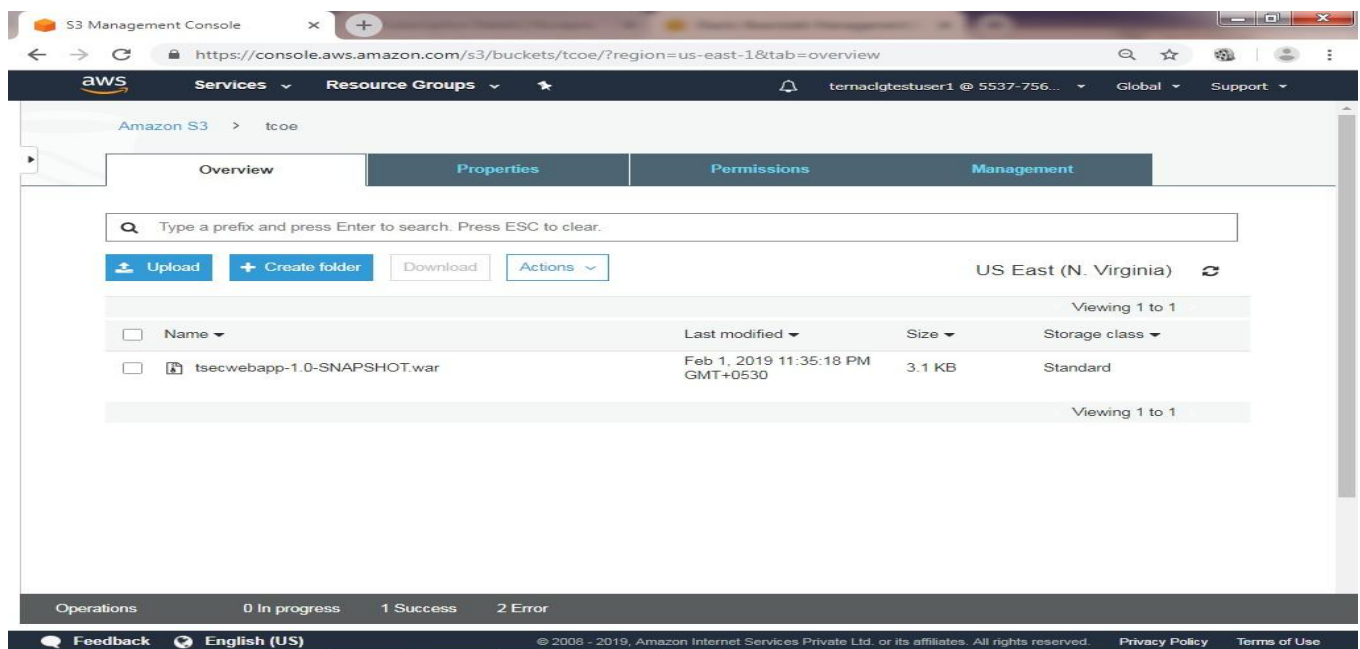
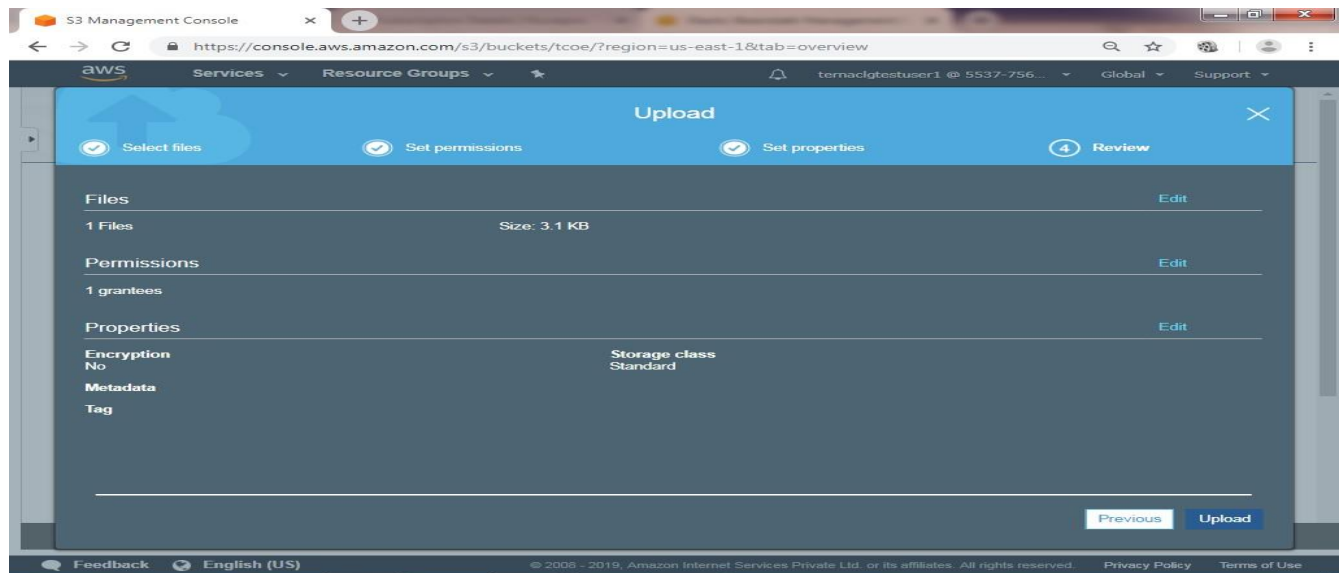
Step 7: Archive Data to Glacier

- Transfer data from S3 to Glacier for long-term archiving.
- Configure Glacier retrieval settings based on data retrieval urgency.

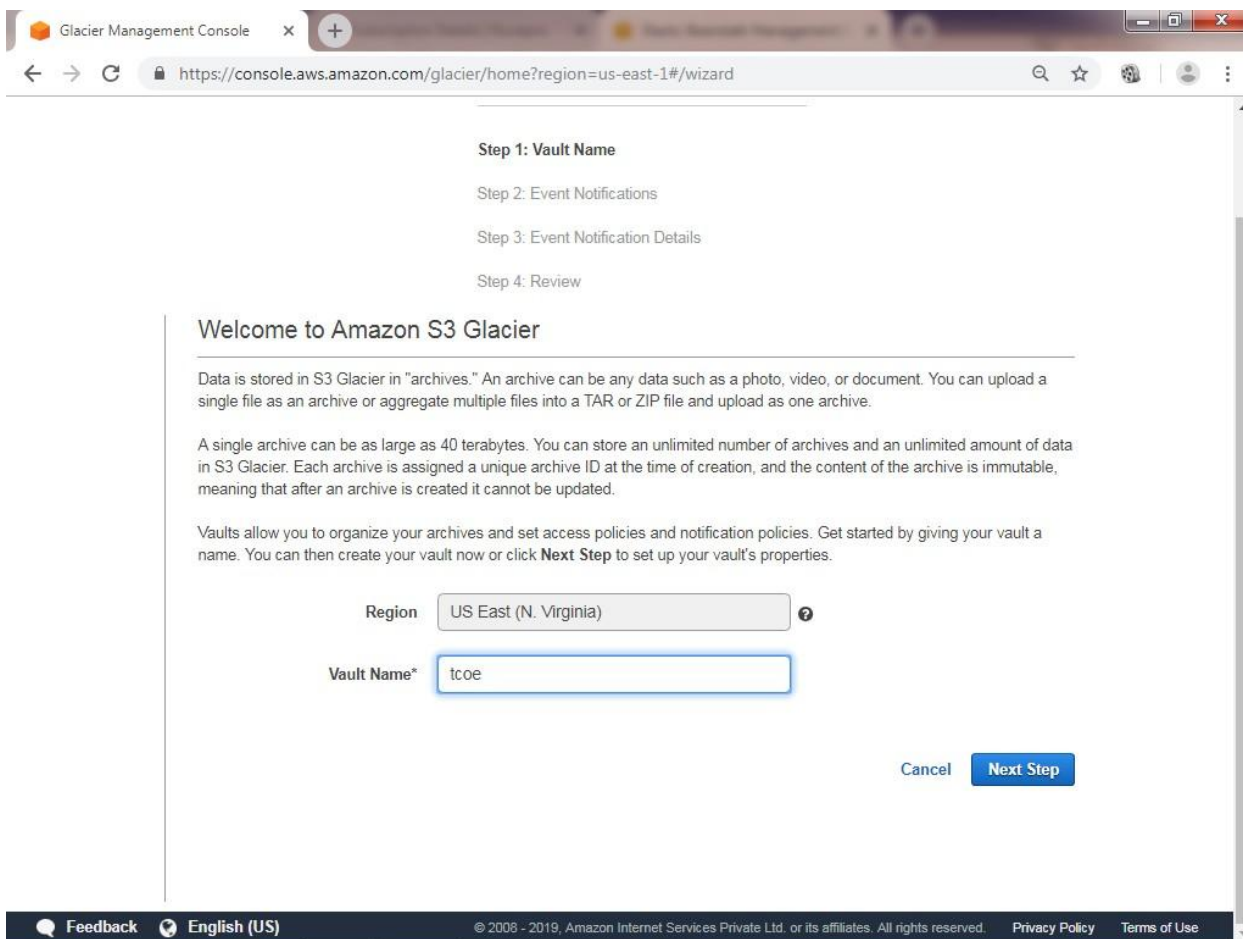
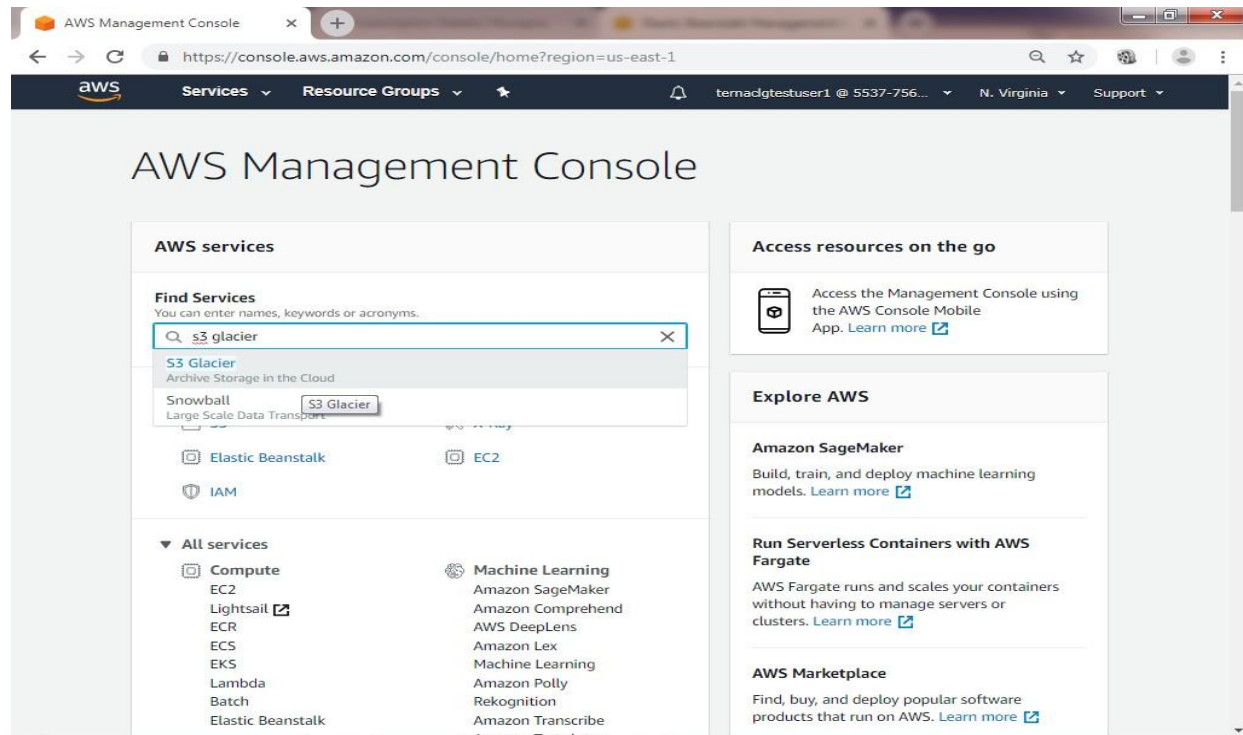








S3 Glaciers



Glacier Management Console

https://console.aws.amazon.com/glacier/home?region=us-east-1#/vaults

Services Resource Groups

ternadgtester1 @ 5537-756... N. Virginia Support

Amazon S3 Glacier Vaults

Create Vault Delete Vault Settings

Filter By Name:

Name	Inventory Last Updated	Size (as of last inventory)	# of Archives (as of last inventory)
tcoe	Not updated yet	--	--

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Create Vault Delete Vault Settings

Filter By Name:

Name	Inventory Last Updated	Size (as of last inventory)	# of Archives (as of last inventory)
tcoe	Not updated yet	--	--

Vault Name: tcoe

Details Notifications Permissions Vault Lock Tags

Region: US East (N. Virginia)

Created on: Feb 1, 2019 11:42:08 PM

Vault ARN: arn:aws:glacier:us-east-1:553775641836:vaults/tcoe

Inventory Last Updated: Not updated yet

Vault Details as of the last inventory update:

Size: --

of Archives: --

S3 Management Console

https://s3.console.aws.amazon.com/s3/home?region=us-east-1#

Services Resource Groups

ternadgtester1 @ 5537-756... Global Support

Create bucket

1 Name and region 2 Configure options 3 Set permissions 4 Review

Name and region

Bucket name: tcoebucket Region: US East (N. Virginia)

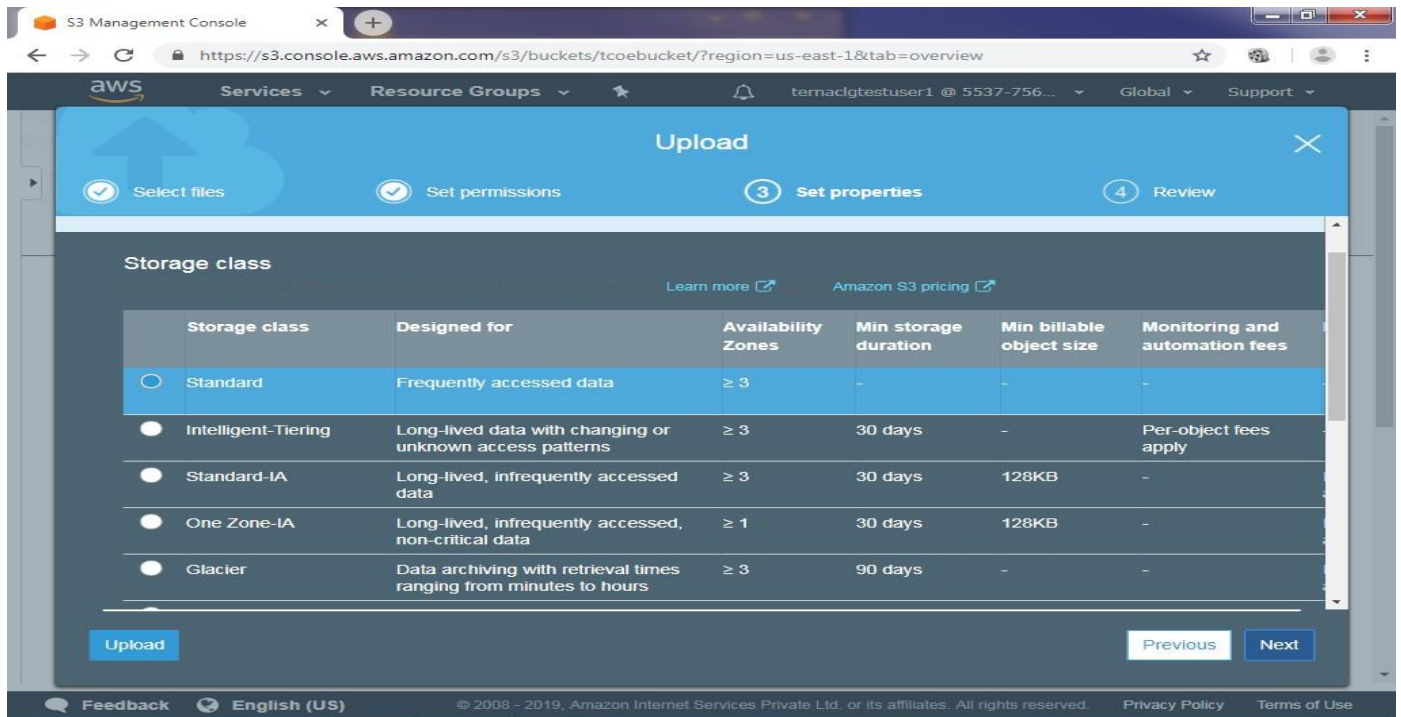
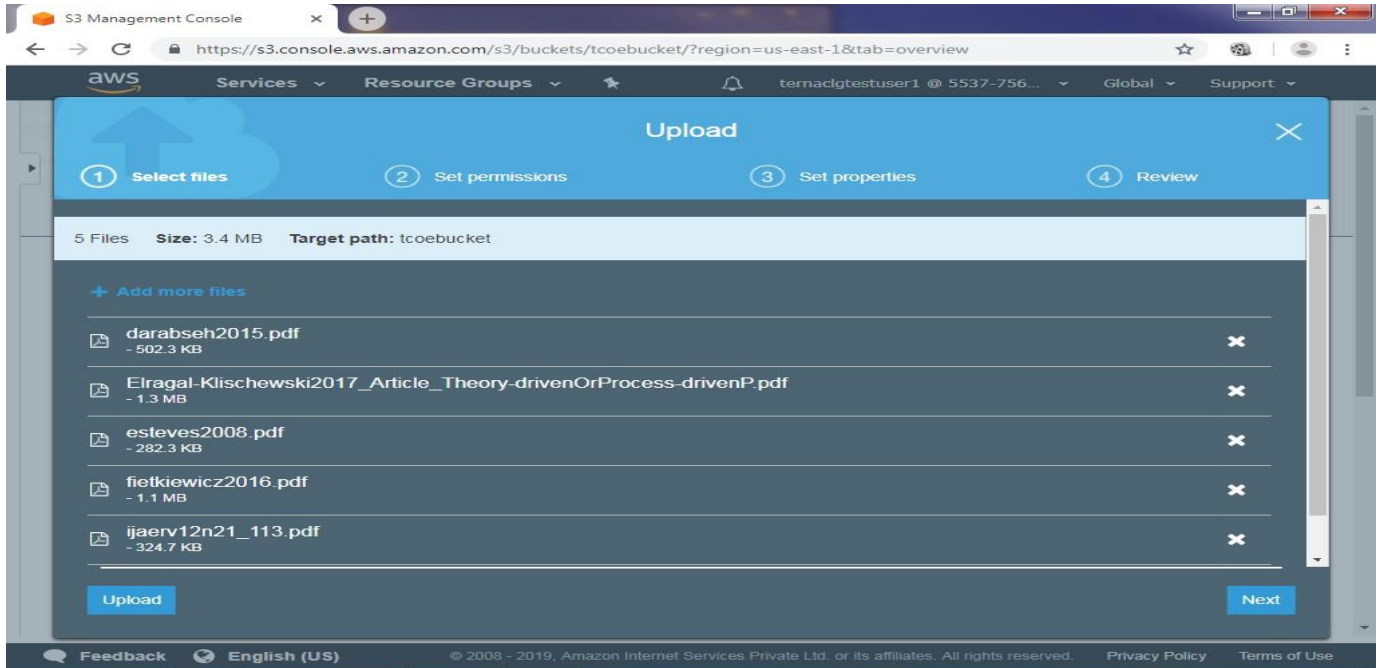
Options

Versioning	Disabled
Server access logging	Disabled
Tagging	0 Tags
Object-level logging	Disabled
Default encryption	None
CloudWatch request metrics	Disabled
Object lock	Disabled

Permissions

Previous Create bucket

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Glacier Management Console

https://console.aws.amazon.com/glacier/home?region=us-east-1#/wizard

Step 1: Vault Name

Step 2: Event Notifications

Step 3: Event Notification Details

Step 4: Review

Welcome to Amazon S3 Glacier

Data is stored in S3 Glacier in "archives." An archive can be any data such as a photo, video, or document. You can upload a single file as an archive or aggregate multiple files into a TAR or ZIP file and upload as one archive.

A single archive can be as large as 40 terabytes. You can store an unlimited number of archives and an unlimited amount of data in S3 Glacier. Each archive is assigned a unique archive ID at the time of creation, and the content of the archive is immutable, meaning that after an archive is created it cannot be updated.

Vaults allow you to organize your archives and set access policies and notification policies. Get started by giving your vault a name. You can then create your vault now or click **Next Step** to set up your vault's properties.

Region: US East (N. Virginia)

Vault Name*

Cancel Next Step

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Glacier Management Console

https://console.aws.amazon.com/glacier/home?region=us-east-1#/vaults

Amazon S3 Glacier Vaults

Create Vault Delete Vault Settings

Filter By Name:

Name	Inventory Last Updated	Size (as of last inventory)	# of Archives (as of last inventory)
tcoe	Not updated yet	--	--

Vault Name: tcoe

Details Notifications Permissions Vault Lock Tags

Region: US East (N. Virginia)

Created on: Feb 2, 2019 7:23:18 AM

Vault ARN: arn:aws:glacier:us-east-1:553775641836:vaults/tcoe

Inventory Last Updated: Not updated yet

Vault Details as of the last inventory update:

Size: --

of Archives: --

Conclusion:

Thus, we have successfully understood and implemented Storage as a Service using AWS S3 and Glacier.